

Genome-first screening of aging and neurodegeneration gene candidates in *Turritopsis dohrnii*

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Abstract - *Turritopsis dohrnii* can reverse from a medusa to a polyp, but genome similarity alone cannot identify the cause. We searched two *T. dohrnii* assemblies and three cnidarian comparators for preselected human aging and neurodegeneration genes with a fixed translated-homology pipeline. Twenty of 24 aging and 16 of 24 neurological genes had strong loci in the main assembly. On an annotated *Clytia* calibration genome, queries from several cnidarians plus humans yielded strong hits for 35 of 47 distinct panel genes versus 27 of 47 using human queries alone. This is a difference in detected-locus yield, not independently measured biological recall. Later whole-proteome, PacBio and RNA checks did not validate a new aging-related gene-family expansion, an extra ATG5 copy, or stage-dependent expression. The output is a bounded coordinate and uncertainty map for future experiments, not a rejuvenation mechanism.

Introduction - The reported ability of *T. dohrnii* medusae to return to the polyp stage motivates tests of genome maintenance and neural-plasticity hypotheses [1]. Prior groups published its genome, high-confidence gene models and comparative orthogroups [2,3]. We therefore ask a narrower question: under a fixed search, which selected human aging and neurodegeneration-gene labels yield candidate homologous genomic loci, and how robust are the resulting copy signals? A locus match does not itself establish orthology or conserved function. The separately published Kyushu gene models and independent Oviedo assembly [2,4] make useful challenges: if a mapping call fails against a second assembly or provider-predicted model, the reason must be investigated rather than silently promoted into species biology. Likewise, *T. rubra* is a valuable near comparator, but two related species cannot isolate lifespan causation.

Methods - Design And Material - Before outcome searches, the expansion fixed two 24-gene panels, H1-H6 and correction families in its repository [4]. Inputs were two *T. dohrnii* assemblies (GCA_027922465.2; GCA_025167195.1), *T. rubra* (GCA_039566895.2), *Aurelia aurita* (GCA_004194415.1), and annotated *Clytia hemisphaerica* (GCF_902728285.1). Exact-gene-verified human and cnidarian query proteins were searched with tBLASTn, e-value at most 1e-5. High-scoring pairs for each label within 50 kb on one contig were grouped; a candidate locus was called strong at 100 summed bits. Calls are homology-search signals, not gene models. Frozen controls included shuffled queries and comparison with earlier panel calls. Exploratory validation compared published Kyushu and *T. rubra* predicted proteomes using low-memory MMseqs; a frozen top 20 was checked against Oviedo *T. dohrnii* and *Clytia* predicted proteins, with Benjamini-Hochberg adjustment over every searched cluster. Full public PacBio DRR267480 was mapped to one whole-genome minimap2 index for a predeclared two-flank ATG5

test. A one-percent RNA pilot sampled one run per stage and counted mapped read ends in target-enriched windows; it cannot support differential-expression inference. Code, locks and numeric ledgers are archived in [4]. The two species-level hypotheses and controls were analyzed under the preregistered H1-H6 family; the later panel-free and structure tests were separately locked before their outcomes. This distinction prevents a post-hoc result from acquiring the status of a preregistered discovery. The follow-up tested two proposed ATG5 regions of 267 bases each and required at least two independent primary molecules with MAPQ at least 30 that span both five-kilobase unique flanks at each region. A locus-crossing read alone cannot pass this rule.

Methods - Ai Assistance Disclosure (Review Required) - AI assistance was used extensively in study planning, code generation, analysis troubleshooting, literature/source triage and manuscript drafting. A signed-in ChatGPT Free account was used for critique and redirection, and Gemini was used in supplementary critique; their responses did not authorize changes to locked outcomes. Automated assistant systems also produced substantial prose, scripts and result summaries. Known verbatim judge prompts and responses are archived in [4], docs/X-JUDGE-ROUNDS.md and docs/X-JUDGE-PROMPTS.md. Verbatim external judge prompts are archived in [4]; the complete prompt/tool/version inventory for all AI-assisted study and writing turns remains to be verified before submission. No AI is listed as an author or cited as a scientific source. The human author must inspect and take responsibility for every claim.

Results - The main *T. dohrnii* assembly gave strong loci for 20/24 aging and 16/24 neurological labels [4]. This is conditional on preselected, conserved-gene panels and is not a species-specific enrichment claim. A paired *Clytia* calibration yielded 35/47 distinct labels with human-plus-cnidarian queries versus 27/47 with human-only queries; the eight extra calls are not verified true-positive recoveries. The negative shuffled-query control produced no strong loci in the five initial genomes. The earlier positive panel control reproduced 52/60 calls, with eight gains from the expanded query set and no losses [4]. The preregistered aging-panel copy contrast had no gene surviving multiple-test correction; the species-separation statistic was -0.013 (exact p=0.5) [4]. These negatives bound interpretation of the positive locus map. The mapped aging fraction is high relative to the scrambled-sequence control, but that control asks whether random protein strings create false hits, not whether the selected aging labels are unusually represented among real conserved genes. A separate 23-gene non-aging decoy yielded strong hits in 8-10 genes depending on genome [4], showing genuine background conservation. A fragmented-genome simulation also lost 2 of 39 detectable genes, so non-detection in a poorer assembly cannot establish evolutionary loss [4].

Exploratory Validation And Discussion - The all-protein *T. dohrnii*/*T. rubra* sweep produced 24,270 provisional similarity clusters; all frozen top-20 clusters had all-cluster adjusted q=1.0. Different annotation totals and unverified orthology prevent a new family-expansion claim [4]. In all 1,704,140 public PacBio subreads, 12 and 27 primary MAPQ60 reads crossed the two putative cnidarian-type

ATG5 spans, but zero at either span covered at least 5 kb of unique sequence on BOTH flanks. Consecutive read ordinals could not establish independent molecules; the proposed extra ATG5 copy remains unresolved, not disproved [4]. Nine paired stage RNA runs share one BioSample accession per stage, and no primary run-to-distinct-extraction crosswalk was verified; the sparse target-window pilot cannot establish stage-specific ATG5 expression [4]. A clean-clone source/hash and one-query BLAST smoke passed, not full raw-data reproduction. Kyushu's independent gene models and earlier OrthoFinder analysis [2] rule out a claim of first annotation; standard orthogroup and family-count methods already exist [3,5]. The useful result is a checkable set of candidate coordinates and explicit evidence grades, rather than a single success label. BLAST hits identify sequence similarity; multiple query species improve search sensitivity; independent assembly agreement gives further confidence; but neither a high score nor an a second aligner searching the same assembly resolves a duplication. Published *T. dohrnii* and *T. rubra* proteomes differ strongly in prediction totals, which can distort family-count tests. Within this limit, the panel-free $q=1.0$ result is a true failure of its frozen enrichment gate, not evidence that all species differ only trivially. The PacBio zero-flank result is a failure of coverage at the required molecular scale, not a proof that the extra ATG5 sequence is an artifact. Both kinds of negative identify exactly what independent evidence a later experiment needs. Future work requires independent molecular structure, specimen-grounded whole-transcriptome counts, and a portable full raw pipeline before any mechanistic conclusion. A future specimen-grounded comparison should count whole-transcriptome fragments against a vetted gene model and compare normal polyp, medusa, cyst and reversed polyp without assuming that multiple runs sharing a BioSample are independent animals. A 2025 preprint already profiled these four stages for selected gene families [6], so a new test would require an independently validated, prespecified contrast rather than repeating its stage plots. A future structural study should use phased long molecules or a high-contiguity independently produced assembly. Until those checks pass, the genomic atlas supports follow-up choices, not biomarkers of human aging.

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